
DROP-OFF & DID NOT ATTEND (DNA) RATE ANALYSIS

(ANA-2)

Dataset Used: NHS Talking Therapies Monthly Statistics
Prepared By: Priyansi Pandey (Data Analytics Intern)
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Project: AI Mental Health Assistant

DROP-OFF & DID NOT ATTEND (DNA) RATE ANALYSIS

1. OBJECTIVE

The purpose of this analysis was to evaluate patient retention and appointment attendance patterns within NHS Talking Therapies services using the March 2026 activity dataset.

The task focused on:

- Calculating Did Not Attend (DNA) rates
- Measuring treatment drop-off across referral stages
- Identifying regions with higher attrition
- Visualizing patient retention through a funnel chart

The analysis was performed using Python with Pandas, Matplotlib, and Seaborn.

2. DATASETS USED:

Dataset:
NHS Digital NHS Talking Therapies Monthly Statistics Including Employment Advisors — March 2026

The dataset contains regional NHS Talking Therapies operational metrics including:

- Referrals received
- Assessments completed
- Treatment sessions attended
- Appointments missed (DNA)
- Completed treatment courses

3. DATA PREPARATION:

The raw dataset initially contained:

- 280,581 rows
- 10 columns

Only the required analytical fields were retained for this task:

Selected Field	Purpose
ORG_NAME1	NHS Region
MEASURE_NAME	Operational metric
MEASURE_VALUE_SUPPRESSED	Metric value

The dataset was then cleaned by:

- Converting numeric fields properly
- Removing suppressed or missing values
- Filtering only regional-level records
- Selecting operational metrics relevant to attendance and retention analysis

4. METRICS COMPUTED:

The following KPIs were engineered during the analysis:

KPI	Formula
DNA Rate (%)	$\text{DNA Appointments} \div (\text{Attended} + \text{DNA Appointments}) \times 100$
Referral-to-Assessment Drop-off (%)	$100 - (\text{First Assessments} \div \text{Referrals Received} \times 100)$
Assessment-to-Second Treatment Drop-off (%)	$100 - (\text{Second Treatments} \div \text{First Assessments} \times 100)$

5. REGIONAL FUNNEL ANALYSIS:

The funnel analysis tracked patient progression through four stages:

1. Referrals Received
2. First Assessment
3. Second Treatment Session
4. Finished Course of Treatment

The North West region was selected as the visualization hotspot because it showed strong referral volume and clear retention decline across treatment stages.

6. KEY FINDINGS:

A. DNA Rates Across Regions:

The analysis showed noticeable differences in appointment attendance behaviour across NHS regions.

Observations:

- DNA rates remained relatively stable across most regions
- Regions with larger referral volumes also showed higher missed appointment counts
- Missed appointments create additional pressure on therapy service capacity and waiting times

Interpretation:

Higher DNA rates may indicate:

- Long waiting periods
- Reduced patient engagement
- Scheduling barriers
- Accessibility challenges

B. Referral-to-Assessment Drop-off:

A significant portion of referrals did not progress to the first clinical assessment stage.

Possible Causes:

- Delayed scheduling
- Patients withdrawing before assessment
- Incomplete referrals
- Capacity limitations within services

This stage showed the largest reduction in patient volume across the treatment pathway.

C. Assessment-to-Treatment Drop-off:

The second major reduction occurred between:

- Initial assessment
- Second treatment session

This indicates that a subset of patients disengage shortly after entering treatment services.

Possible contributors include:

- Therapy dissatisfaction
- Improvement before treatment continuation
- Personal or logistical barriers
- Service waiting delays

7. FUNNEL CHART INTERPRETATION:

The funnel chart visually demonstrated progressive patient reduction across treatment stages.

Key Insight

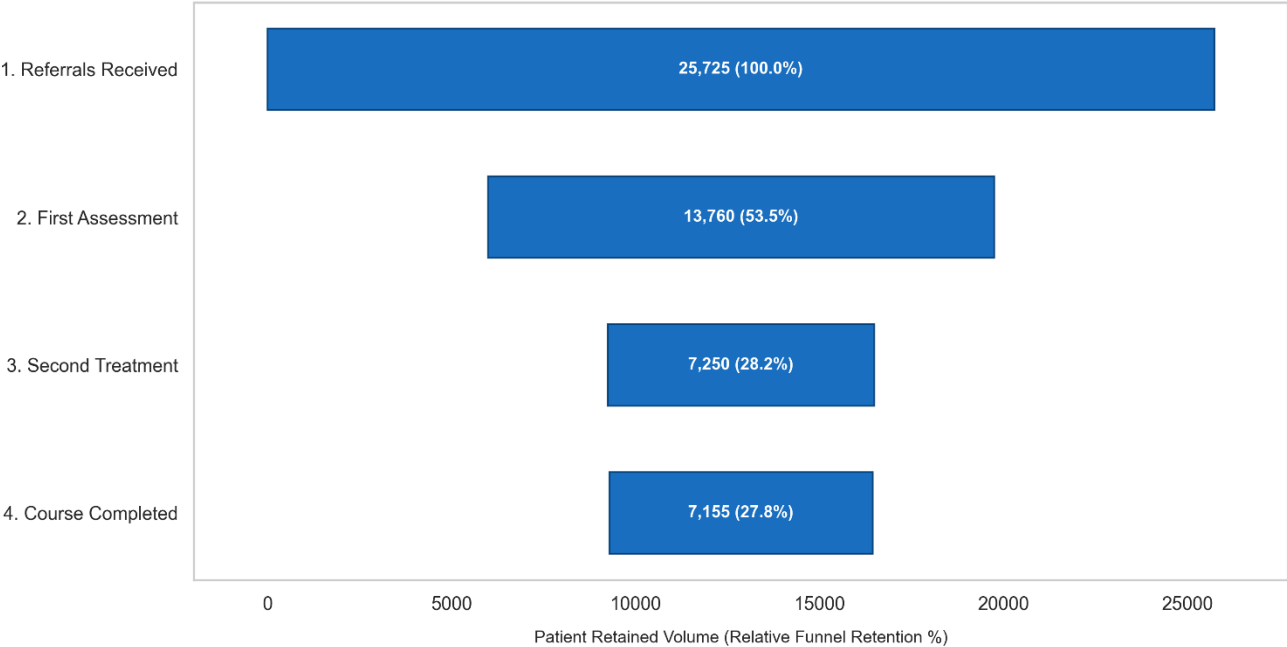
The steepest decline occurred between:

- Referral received
- First assessment

This suggests that early-stage patient retention is the primary operational bottleneck.

The chart also showed that only a fraction of the original referrals ultimately completed treatment pathways.

Figure 7: Clinical Patient Pathway Funnel & Retention Rates - North West Hotspot (March 2026)



8. OUTPUT FILES GENERATED:

The following deliverables were successfully generated from the pipeline:

File	Description
iapt_funnel_dropoff_matrix.csv	Regional KPI matrix
iapt_treatment_funnel_chart.png	Funnel retention visualization

9. TECHNOLOGIES USED:

Tool	Purpose
Python	Data analysis
Pandas	Data cleaning and transformation
Matplotlib	Funnel chart generation
Seaborn	Visualization styling

10. CONCLUSION:

This analysis successfully evaluated operational retention performance within NHS Talking Therapies services using March 2026 regional activity data.

The findings indicate that:

- Patient attrition begins early in the referral pipeline
- Missed appointments remain a measurable operational issue
- Treatment continuation declines progressively across therapy stages

The generated funnel analysis provides a useful operational view of service retention and can support future work related to:

- waiting time optimisation
- engagement improvement
- regional benchmarking
- predictive dropout modelling